where the matrix $\boldsymbol{B}(z) = (b_{ij}(z))$ is related to $\boldsymbol{A}(z)$ by

$$\boldsymbol{B}(z) = \boldsymbol{\Phi}'^{T}(z)\boldsymbol{A}(\boldsymbol{\Phi}(z))\boldsymbol{\Phi}'(z).$$

We say that a 2-form ψ is *conserved under mapping* $\boldsymbol{\Phi}$ if

$$\boldsymbol{\Phi}^{*}\psi = \psi.$$

In coordinates, the conservation of the 2-form represented by matrix $\boldsymbol{A}$ under a mapping $\boldsymbol{\Phi}$ means that

$$\boldsymbol{\Phi}'^{T}\boldsymbol{A}\boldsymbol{\Phi}' = \boldsymbol{A}.$$

In the particular case of the symplectic 2-form ψ_{S}, we have $\boldsymbol{A} = \boldsymbol{J}$, and the following condition for conservation under the mapping $\boldsymbol{\Phi}$

$$\boldsymbol{\Phi}'^{T}\boldsymbol{J}\boldsymbol{\Phi}' = \boldsymbol{J}. \tag{2.17}$$

A map that conserves the symplectic 2-form, or, in coordinates, satisfies (2.17), is termed a *symplectic map*.

Taking the determinant of both sides of (2.17), we have

$$\det(\boldsymbol{\Phi}'^{T}\boldsymbol{J}\boldsymbol{\Phi}') = \det(\boldsymbol{J}) \Rightarrow \det(\boldsymbol{\Phi}'^{T})\det(\boldsymbol{J})\det(\boldsymbol{\Phi}') = \det(\boldsymbol{J}),$$

hence

$$\det(\boldsymbol{\Phi}')^{2} = 1,$$

so $|\det(\boldsymbol{\Phi}')| = 1$. This raises a curious issue since we could conceivably have $\det(\boldsymbol{\Phi}') = -1$.

In the case of a flow map $\boldsymbol{\Phi} = \mathscr{F}_t$, we know that for $t \to 0$, the map reduces to the identity map (the same has to happen for a consistent numerical method). This means that we would have to have

$$\lim_{t\to 0} \det(\mathscr{F}_t') = 1.$$

If the system is Hamiltonian, the map is symplectic for all t, and the determinant will be a continuous function of t, so the cases of interest have $\det(\mathscr{F}_t') = +1$. The flow map of a Hamiltonian system is volume preserving.